

# AlgoStalk — The VAR/VECM Family: Complete YouTube Narration Script

Hook (Option A) + Full Narration + All 17 Expansion Points Integrated

Slides renumbered sequentially | [NEW] = expansion content | [HOOK] = hook content

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**HOW TO USE:** Every [SLIDE X] is one slide. Build your deck from these in order. Below each slide number: the **visual description** first, then the **narration**. [NEW] marks slides added from the expansion. [HOOK] marks the opening hook. Running example throughout: **Gold (XAU), US Dollar Index (DXY), S&P 500**.

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## PART 0 — THE HOOK [HOOK]

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[SLIDE 1] [HOOK] *Visual: Split screen. LEFT: a person confidently typing in front of an ARIMA chart. RIGHT: a giant flashing red loss on a trading screen. No text. No music. Hold for 3 seconds.*

*(cold open — silence for 2 seconds)*

You built the model. You checked the assumptions. You ran the forecast.

And then the market did something your model had absolutely no idea was coming.

Not because you made a mistake.

Because you used a tool that was only ever designed to look at ONE thing.

While the answer was hiding in the relationship between THREE things.

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[SLIDE 2] [HOOK] *Visual: Three lines — Gold, Dollar Index, S&P 500 — all on one chart. They are clearly reacting to each other. Arrows between them pulsing. Text at bottom: "They talk to each other. Is your model listening?"*

Right now, Gold is reacting to the Dollar.

The Dollar is reacting to the stock market.

And the stock market is reacting to Gold.

All three. At the same time. In real time.

If your model is only watching one of them — you are sitting in an interrogation room interviewing only one witness while two others are sitting right outside.

You are going to get an incomplete confession.

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**[SLIDE 3] [HOOK]** *Visual: The detective walking into a room with three suspects. Confident stride. The three suspects labeled: GOLD / DXY / S&P500. The detective pulls out a case file labeled "VAR/VECM."*

Today we fix that.

In this video, you are going to learn the entire VAR and VECM family — the most powerful set of multivariate detective tools in all of quantitative finance.

And I am not going to give you a surface level tour.

By the time this video ends, you will be able to:

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**[SLIDE 4] [HOOK]** *Visual: Five bullet points that TYPE onto screen one by one as you say each. Bold. Fast.*

- Identify which markets lead and which markets follow
- Calculate exactly how a shock in one market travels to all others
- Quantify who is responsible for tomorrow's uncertainty
- Detect long-run bonds between markets that ARIMA can never see
- Model markets that switch between calm and crisis modes automatically

Not conceptually. Not approximately. Calculated. With real numbers. Step by step.

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**[SLIDE 5] [HOOK]** *Visual: Rapid-fire preview montage — 6 still frames cycling fast (like a movie trailer): Frame 1: VAR equation system — three variables connected Frame 2: IRF plot showing ripple effect of a shock Frame 3: Cholesky triangle with numbers filled in Frame 4: FEVD pie chart splitting market drivers Frame 5: VECM equilibrium error bouncing around zero Frame 6: Markov-Switching regime probability plot spiking during 2008 (each frame 0.5 seconds, then freeze on AlgoStalk detective logo)*

This is not a beginner video.

This is a complete case file — from the first equation to the most advanced tools used by quantitative analysts at hedge funds and central banks.

And we are going to explain every single one of them like you are a junior detective who has never seen them before.

Because you do not need a PhD to understand these tools. You need a good teacher and enough patience to follow the clues.

These exact tools — VAR, VECM, SVAR — appear in published research from the Federal Reserve, the Bank of England, and JPMorgan's macro strategy desk. Today we learn how

they actually work.

*(brief pause)*

Let us go.

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## PART 1 — THE LIMITATION

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[SLIDE 6] *Visual: One blinking cursor. The word "SOLVED?" fades in with a question mark.*

You have been doing good work so far.

You learned ARIMA. You can handle trending data. You can remove seasonality. You can forecast a single series.

You learned GARCH. You understand that volatility is not constant. You can model fear. You can forecast risk.

You are good. I will give you that.

But today I am going to expose a gap in your detective skills that most people never notice until it is too late.

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[SLIDE 7] *Visual: A crime scene with THREE suspects, but only ONE detective's file open. The other two files are blank.*

Here is the gap.

Everything you have learned so far is about ONE variable.

ARIMA models one variable over time. GARCH models the volatility of one variable over time.

But the real world never gives you one variable. The real world gives you Gold. AND the US Dollar. AND the Stock Market. AND Oil. AND Interest Rates. All at the same time. All connected to each other. All influencing each other.

And here is the uncomfortable truth: if you model them one at a time, you are ignoring the most important information.

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[SLIDE 8] *Visual: Three columns: GOLD / DXY / S&P 500 — each with a zigzag line. Arrows pointing between them with question marks.*

Today's case has three variables.

**Gold** — XAU/USD. The safe haven asset. People run to Gold when they are scared.

**The US Dollar Index** — DXY. Measures the dollar's strength against a basket of currencies. Gold and the Dollar have a famously complicated relationship.

**The S&P 500** — The mood ring of the American economy.

These three do not live in separate rooms. They live in the same house. They hear everything. They react to everything.

The question is: who is reacting to whom? Who leads? Who follows? And who is causing what?

That is exactly what the VAR family was built to answer.

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[SLIDE 9] *Visual: Title card: "THE VAR/VECM FAMILY" — detective board style with strings connecting all model names: VAR → IRF → Granger Causality → GIRF → FEVD → Cointegration → VECM → SVAR → Structural FEVD → Time-Varying VAR → Markov-Switching VAR*

Welcome to the most powerful family of tools in multivariate time series analysis.

By the end of this video, you will have the full picture. Every tool. Every idea. Every connection.

AlgoStalk. Let us begin.

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## CHAPTER 1 — WHAT IS VAR?

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[SLIDE 10] *Visual: The word VAR. Below it: "Vector AutoRegression." Three words, each underlined separately.*

Vector AutoRegression. Three words. Let me break each one down.

**Auto** — the model uses its own past values to predict itself. You know this from ARIMA.

**Regression** — it uses a regression equation. Also familiar.

**Vector** — a vector means multiple things at once. Not one series. A whole group of series, stacked together.

So VAR is: multiple time series, all regressing on each other's past values at the same time.

That is it. That is the entire idea.

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[SLIDE 11] *Visual: AR(1) equation on the left. VAR(1) system of three equations on the right. Highlight the structural difference.*

You remember the AR(1) model. One variable. One equation. Yesterday's value of Y predicts today's Y.

Now imagine we have three variables: Gold, Dollar, and S&P 500.

VAR says: write one equation for EACH of them. And in each equation, use the PAST VALUES OF ALL THREE as predictors.

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**[SLIDE 12]** *Visual: Three VAR equations written out cleanly.*

$$\begin{aligned}\text{Gold}_t &= c_1 + a_{11} \cdot \text{Gold}_{t-1} + a_{12} \cdot \text{DXY}_{t-1} + a_{13} \cdot \text{SP500}_{t-1} + \varepsilon_{1t} \\ \text{DXY}_t &= c_2 + a_{21} \cdot \text{Gold}_{t-1} + a_{22} \cdot \text{DXY}_{t-1} + a_{23} \cdot \text{SP500}_{t-1} + \varepsilon_{2t} \\ \text{SP500}_t &= c_3 + a_{31} \cdot \text{Gold}_{t-1} + a_{32} \cdot \text{DXY}_{t-1} + a_{33} \cdot \text{SP500}_{t-1} + \varepsilon_{3t}\end{aligned}$$

Look carefully. Every variable talks to every other variable. No variable is isolated.

Equation 1 says: today's Gold depends on yesterday's Gold, yesterday's Dollar, AND yesterday's S&P 500.

This is VAR(1) — because we look back one period.

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**[SLIDE 13]** *Visual: The 3×3 coefficient matrix A highlighted. All nine elements labeled  $a_{11}$  through  $a_{33}$ .*

Count the parameters: three equations, three coefficients for lagged variables plus one constant each. That is 12 parameters. For just ONE lag.

Add a second lag — VAR(2) — and you have 24 parameters.

VAR eats parameters quickly. More variables, more lags, more parameters. More parameters need more data. Always.

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**[SLIDE 14]** *Visual: Compact matrix notation:  $Y_t = c + A_1 \cdot Y_{t-1} + \varepsilon_t$*

In matrix notation, the whole system compresses beautifully.

$Y_t$  equals  $c$  plus  $A_1$  times  $Y_{t-1}$  plus  $\varepsilon_t$ .

$Y$  is a vector of all three variables.  $A_1$  is the 3×3 coefficient matrix. Epsilon is a vector of three error terms.

For VAR( $p$ ), we just add more matrices:  $A_1$  through  $A_p$ .

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**[SLIDE 15]** *Visual: The error covariance matrix Sigma — 3×3 with off-diagonal elements highlighted in color.*

Very important — the error terms in VAR are NOT independent of each other.

If something unexpected shocks Gold today... that same event probably also shocked the Dollar and the S&P 500 on the same day.

Think about a surprise geopolitical crisis. Gold spikes. Dollar moves. Stocks fall. All three received a shock at the same moment.

We capture this with the variance-covariance matrix of errors — Sigma. Diagonal elements are variances of each error. Off-diagonal elements are covariances between errors.

This matrix becomes very important when we reach Orthogonalized IRF. Hold that thought.

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**[SLIDE 16]** *Visual: Table with 5 rows of data: Date, Gold, DXY, S&P500.*

| Date  | Gold  | DXY   | S&P500 |
|-------|-------|-------|--------|
| ----- | ----- | ----- | -----  |
| Day 1 | 1800  | 103.2 | 4200   |
| Day 2 | 1812  | 102.8 | 4215   |
| Day 3 | 1795  | 103.5 | 4198   |
| Day 4 | 1820  | 102.6 | 4240   |
| Day 5 | 1808  | 103.0 | 4225   |

Let us make this concrete. Five days of data. Three variables.

To estimate VAR(1), we run OLS separately on each equation. For Equation 1, regress today's Gold on yesterday's Gold, yesterday's DXY, and yesterday's S&P 500. Standard regression. You already know how to do this.

Do the same for Equations 2 and 3. Three regressions. Done.

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**[SLIDE 17]** *Visual: SUR property — because all equations share the same regressors, OLS on each gives identical efficiency to joint estimation.*

Because all three equations have the exact same set of regressors, running OLS on each equation separately gives the exact same result as running them jointly. Statisticians call this SUR — Seemingly Unrelated Regressions.

Separate regressions. Same efficiency as a joint system. That is a detective who got lucky — the tools already in the kit were enough.

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## CHAPTER 2 — CORRELATED ERRORS: PROOF WITH NUMBERS [NEW]

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**[SLIDE 18] [NEW]** *Visual: Table of 6 days of VAR residuals. Two columns: Gold residual ( $\varepsilon_1$ ) and DXY residual ( $\varepsilon_2$ ).*

| Day | Gold Residual ( $\varepsilon_1$ ) | DXY Residual ( $\varepsilon_2$ ) |
|-----|-----------------------------------|----------------------------------|
| 1   | +0.50%                            | -0.40%                           |
| 2   | -0.30%                            | +0.20%                           |
| 3   | +0.80%                            | -0.65%                           |
| 4   | -0.20%                            | +0.15%                           |
| 5   | +0.40%                            | -0.35%                           |
| 6   | -0.60%                            | +0.48%                           |

Now let me show you concretely how we know errors are correlated. After fitting the VAR, we extract the residuals — the part of each variable our model could not explain. Six observations, two variables.

Look at this table before doing any math.

Day 1: Gold surprised positively. DXY surprised negatively. Day 2: Gold negative, DXY positive. Day 3: Gold positive, DXY negative.

Every time Gold surprised to the upside, DXY surprised to the downside ON THE SAME DAY. You can SEE the negative correlation before touching a calculator.

This simultaneous opposite movement is exactly what correlated residuals look like. And it is why we cannot just shock "DXY alone" — a real DXY shock arrives with a correlated Gold shock on the same trading day.

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**[SLIDE 19] [NEW]** *Visual: Step-by-step covariance calculation. Each term of the product sum filled in from the table.*

Now we make it precise.

### Step 1: Means.

Mean of  $\varepsilon_1 = (0.50 - 0.30 + 0.80 - 0.20 + 0.40 - 0.60) / 6 = 0.60 / 6 = +0.10\%$  Mean of  $\varepsilon_2 = (-0.40 + 0.20 - 0.65 + 0.15 - 0.35 + 0.48) / 6 \approx -0.095\%$

Both close to zero. Good — residuals from a properly fitted model should have zero mean.

**Step 2: Sample Covariance =  $(1/(n-1)) \times \sum (\varepsilon_{1i} - \text{mean}_1)(\varepsilon_{2i} - \text{mean}_2)$**

$$\begin{aligned}
\text{Day 1: } & (0.50 - 0.10) \times (-0.40 + 0.095) = 0.40 \times -0.305 = -0.122 \\
\text{Day 2: } & (-0.30 - 0.10) \times (0.20 + 0.095) = -0.40 \times 0.295 = -0.118 \\
\text{Day 3: } & (0.80 - 0.10) \times (-0.65 + 0.095) = 0.70 \times -0.555 = -0.389 \\
\text{Day 4: } & (-0.20 - 0.10) \times (0.15 + 0.095) = -0.30 \times 0.245 = -0.074 \\
\text{Day 5: } & (0.40 - 0.10) \times (-0.35 + 0.095) = 0.30 \times -0.255 = -0.077 \\
\text{Day 6: } & (-0.60 - 0.10) \times (0.48 + 0.095) = -0.70 \times 0.575 = -0.403 \\
& \text{SUM} = -1.183
\end{aligned}$$

Covariance =  $-1.183 / 5 = -0.237$ . Negative. Confirmed in numbers.

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**[SLIDE 20] [NEW]** Visual: Full 2×2 Sigma matrix filled in. Correlation formula below. Final result highlighted.

Our complete error covariance matrix:

$$\text{Sigma} = \begin{vmatrix} 0.040 & -0.012 \\ -0.012 & 0.025 \end{vmatrix}$$

$\text{Var}(\varepsilon_1) = 0.040 \rightarrow$  Gold's typical unexplained daily move:  $\pm 0.20\%$   $\text{Var}(\varepsilon_2) = 0.025 \rightarrow$  DXY's typical unexplained daily move:  $\pm 0.158\%$   $\text{Cov} = -0.012$

Converting to correlation for a standardized number:

$$\text{Correlation} = -0.012 / (0.200 \times 0.158) = -0.38$$

Moderate negative correlation. When Gold gets an unexpected positive shock — say surprise safe-haven buying — DXY tends to receive a negative shock on the same day. Markets rotated INTO gold and OUT of dollar assets simultaneously.

This is exactly why shocking "DXY alone" is unrealistic. This is the entire reason we need Cholesky decomposition. We just proved it with numbers.

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## CHAPTER 3 — STATIONARITY CHECK

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**[SLIDE 21]** Visual: STATIONARITY warning sign. "VAR requires stationary inputs."

Before we estimate any VAR, we need to talk about stationarity.

A stationary series has a constant mean, constant variance, and no trend running away forever. VAR has the same requirement as ARIMA — all variables entering the VAR must be stationary.

Gold prices? Trending. Not stationary. DXY? Not stationary over the long run. S&P 500? Definitely not.

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[SLIDE 22] *Visual: Two charts side by side — Gold price level (trending) and Gold log returns (flat around zero).*

We transform. We take log returns:  $\log(\text{Gold}_t / \text{Gold}_{t-1})$ .

Now Gold returns are stationary — bouncing around a mean. We do the same for DXY and S&P 500.

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[SLIDE 23] *Visual: ADF test result table. Level: Non-Stationary ❌. Log Returns: Stationary ✅.*

We confirm stationarity formally using the Augmented Dickey-Fuller test.

Gold price level: unit root detected. Non-stationary. ❌ Gold log returns: no unit root. Stationary. ✅

Same result for DXY and S&P 500 returns. We proceed with the three return series.

A good detective checks their evidence before using it. This is that check.

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## CHAPTER 4 — LAG SELECTION WITH AIC AND BIC

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[SLIDE 24] *Visual: VAR(1) through VAR(5) in a row with question marks above each.*

How far back should our model look? One lag? Two? Five?

We use AIC and BIC to answer it — exactly as we did for GARCH order selection.

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[SLIDE 25] *Visual: AIC and BIC formulas.*

$$\begin{aligned}\text{AIC} &= -2 \cdot \log(\text{likelihood}) + 2 \cdot k \\ \text{BIC} &= -2 \cdot \log(\text{likelihood}) + k \cdot \log(n)\end{aligned}$$

Both saying: "I want the model that fits the data well, but I penalize every extra parameter you add." AIC penalizes lightly — tends to pick more generous models. BIC penalizes harshly — tends to pick leaner ones.

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[SLIDE 26] *Visual: Lag selection table with asterisks marking best values.*

| Lag | AIC      | BIC      | HQ       |
|-----|----------|----------|----------|
| 1   | -18.421  | -18.203* | -18.335* |
| 2   | -18.467* | -18.109  | -18.328  |
| 3   | -18.431  | -17.933  | -18.232  |

AIC says lag 2. BIC says lag 1. They disagree.

Default: start with BIC's suggestion. Run the model. Check if residuals still show autocorrelation. If they do — the model missed something — move to lag 2.

In our case, VAR(2) cleans up the residuals properly. So we go with lag 2.

[SLIDE 27] *Visual: Ljung-Box test result — residuals pass white noise check.*

After fitting VAR(2), we test residuals with the Ljung-Box test. Good residuals should be white noise — no pattern, no autocorrelation remaining.

Our residuals pass. VAR(2) is our model. The crime scene has been fully processed. No clues left behind.

## CHAPTER 5 — GRANGER CAUSALITY

[SLIDE 28] *Visual: Two magnifying glasses on GOLD and DXY. Arrow between them with question mark.*

We have our VAR(2). Now the first real detective question: who is causing what?

Does the Dollar drive Gold? Or does Gold drive the Dollar? Or is it both ways?

This is the question Granger Causality was designed to answer.

I want to be very clear before we go further: Granger causality is NOT regular causality. It is something more specific and more honest.

[SLIDE 29] *Visual: Definition box — plain English.*

The definition:

**X Granger-causes Y if: adding X's past values to our model makes predictions of Y significantly better than using Y's own past alone.**

Not "X physically causes Y." Just: "X's past contains information about Y's future that Y's own past does not already contain."

It is like saying: "When that witness starts talking, the case moves faster."

Information. Predictive power. That is what Granger causality measures.

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[SLIDE 30] *Visual: Two model equations side by side — restricted (Gold only) and unrestricted (Gold + DXY).*

Restricted:  $\text{Gold}_t = c + a_{1,1}\text{Gold}_{t-1} + a_{1,1}\text{Gold}_{t-2} + \varepsilon_t$

Unrestricted:  $\text{Gold}_t = c + a_{1,1}\text{Gold}_{t-1} + a_{1,1}\text{Gold}_{t-2}$   
 $+ a_{1,2}\text{DXY}_{t-1} + a_{1,2}\text{DXY}_{t-2} + \varepsilon_t$

If the unrestricted model is significantly better — if DXY's past lags are jointly significant — we say: DXY Granger-causes Gold. We measure this with an F-test.

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[SLIDE 31] *Visual: F-test result.*

$H_0$ : DXY does NOT Granger-cause Gold

F-statistic: 6.82      p-value: 0.001

→ REJECT  $H_0$ . DXY Granger-causes Gold. 

The p-value is 0.001. Well below 0.05. The Dollar's past contains real information about Gold's future.

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[SLIDE 32] *Visual: Reversed test — Gold causing DXY.*

$H_0$ : Gold does NOT Granger-cause DXY

F-statistic: 1.21      p-value: 0.298

→ FAIL TO REJECT. Gold does NOT Granger-cause DXY. 

Interesting. Dollar leads Gold. Gold does NOT lead Dollar. The information flows one direction.

In detective terms: DXY is giving testimony that helps solve the Gold case. Gold is not returning the favor.

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[SLIDE 33] *Visual: Full 3×3 Granger causality matrix for all three variables.*

|        | → Gold | → DXY | → S&P500 |
|--------|--------|-------|----------|
| Gold   | —      | ✗     | ✓        |
| DXY    | ✓      | —     | ✓        |
| S&P500 | ✓      | ✗     | —        |

Dollar causes Gold and S&P. S&P and Gold have a two-way relationship. Dollar is upstream — it sends information but does not receive from Gold or stocks.

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[SLIDE 34] *Visual: Big warning sign: "Granger Causality ≠ True Causality"*

Important warning — remember this every time you run a Granger test.

Granger causality tells you about predictive information, not physical cause and effect. Maybe a third variable is driving both Gold and the Dollar. Maybe the relationship is driven by one unusual period.

Granger causality is a clue. A very useful clue. But not the final verdict. Always combine with economic reasoning.

Even Sherlock Holmes did not convict on one clue alone.

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## CHAPTER 6 — IRF: IMPULSE RESPONSE FUNCTION

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[SLIDE 35] *Visual: Three variable nodes with one getting a lightning bolt shock. Ripple effects spreading to others over time.*

Granger causality told us WHO leads. Now we want to know HOW the system actually responds over time.

If today there is a big unexpected shock to the Dollar — say a surprise Fed announcement — what happens? Does Gold fall immediately? Does it take three days? Does it overshoot then correct?

These are the questions Impulse Response Functions were built to answer.

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[SLIDE 36] *Visual: A single shock spike at t=0. A path traced out over time periods 1 through 10 showing the response of another variable.*

An IRF shows: if one variable receives a one-standard-deviation shock at time zero, how does each variable in the system respond over the next N time periods?

It is a time-travel experiment. Imagine everything is calm. One thing shocks. What is the domino effect?

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[SLIDE 37] *Visual: The key clarification — VMA as base, not VAR directly.*

Before we calculate: is IRF based on the VAR equation or something else?

**Everything related to IRF is fundamentally rooted in the VMA representation — not the VAR directly.**

The VAR equation says: "Y today depends on Y yesterday." That is backward-looking. Hard to ask "what happens if I shock Y today" from a backward equation.

The VMA equation says: "Y today = weighted sum of ALL past shocks." That is a forward expression. It naturally answers: "If one shock changes, by how much does Y at each future period change?"

That change per unit shock is the impulse response function.

The process is: Estimate VAR → Convert VAR to VMA → Read off the VMA coefficients — those ARE the impulse responses.

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[SLIDE 38] *Visual: VMA representation equation.*

$$\begin{aligned} Y_t &= \varepsilon_t + A \cdot \varepsilon_{t-1} + A^2 \cdot \varepsilon_{t-2} + A^3 \cdot \varepsilon_{t-3} + \dots \\ &= \sum_{h=0}^{\infty} \Psi_h \cdot \varepsilon_{t-h} \end{aligned}$$

The conversion is elegant. For VAR(1): substitute  $Y_{t-1} = A \cdot Y_{t-2} + \varepsilon_{t-1}$  into the VAR equation. Keep substituting backwards infinitely. You get the VMA.

The result:  $\Psi_h = A^h$ . The impulse response at horizon h IS the coefficient matrix A raised to the power h.

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[SLIDE 39] [NEW] *Visual: Step 1 — shock vector for DXY. Gold gets zero, DXY gets one unit.*

Let us calculate a complete IRF from first principles. Our VAR(1):

$$A = \begin{vmatrix} 0.4 & -0.3 \\ 0.1 & 0.5 \end{vmatrix}$$

We give a one-unit shock to DXY (variable 2) at time  $t=0$ . Gold (variable 1) gets zero shock.

$$\begin{aligned} \text{Shock vector: } \varepsilon_0 &= \begin{vmatrix} 0 \\ 1 \end{vmatrix} \quad \leftarrow \text{Gold: no shock} \\ &\quad \begin{vmatrix} 1 \end{vmatrix} \quad \leftarrow \text{DXY: one unit shock} \end{aligned}$$

**Horizon  $h=0$  (impact period):**

$$\text{Response} = \Psi_0 \times \varepsilon_0 = I \times \varepsilon_0 = [0, 1]$$

Gold response = 0. DXY response = 1.

Of course. A shock lands where it lands. Nothing has propagated yet.

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[SLIDE 40] [NEW] *Visual: h=1 calculation — matrix multiplication  $\Psi_1 \times \varepsilon_0$  shown explicitly.*

**Horizon h=1 (one period later):**

$$\text{Response} = \Psi_1 \times \varepsilon_0 = A \times \varepsilon_0$$

$$\begin{bmatrix} 0.4 & -0.3 \\ 0.1 & 0.5 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.4 \times 0 + (-0.3) \times 1 \\ 0.1 \times 0 + 0.5 \times 1 \end{bmatrix} = \begin{bmatrix} -0.30 \\ 0.50 \end{bmatrix}$$

**Gold at h=1: -0.30.** The DXY shock transmitted to Gold — a -0.30% move. This is the propagation mechanism. The VAR coefficient matrix A is the transmission channel.

**DXY at h=1: +0.50.** DXY continues to feel its own momentum.

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[SLIDE 41] [NEW] *Visual: h=2 calculation —  $A^2$  computed explicitly.*

**Horizon h=2 (two periods later):**

First compute  $A^2$ :

$$\begin{aligned} A^2 &= \begin{bmatrix} 0.4 \times 0.4 + (-0.3) \times 0.1 & 0.4 \times (-0.3) + (-0.3) \times 0.5 \\ 0.1 \times 0.4 + 0.5 \times 0.1 & 0.1 \times (-0.3) + 0.5 \times 0.5 \end{bmatrix} \\ &= \begin{bmatrix} 0.16 - 0.03 & -0.12 - 0.15 \\ 0.04 + 0.05 & -0.03 + 0.25 \end{bmatrix} = \begin{bmatrix} 0.13 & -0.27 \\ 0.09 & 0.22 \end{bmatrix} \end{aligned}$$

Response =  $A^2 \times \varepsilon_0$ :

$$\begin{bmatrix} 0.13 & -0.27 \\ 0.09 & 0.22 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.27 \\ 0.22 \end{bmatrix}$$

**Gold at h=2: -0.27.** Still reacting, still negative, gradually decaying. **DXY at h=2: +0.22.** Also decaying — momentum fading.

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[SLIDE 42] [NEW] *Visual: Full IRF table — Gold and DXY responses to a DXY shock.*

| Horizon h | Gold response | DXY response |
|-----------|---------------|--------------|
| -----     | -----         | -----        |
| 0         | 0.000         | 1.000        |
| 1         | -0.300        | 0.500        |
| 2         | -0.270        | 0.220        |
| 3         | -0.174        | 0.083        |
| 4         | -0.095        | 0.024        |

Reading the story: a one-unit DXY surprise at time zero. Gold feels nothing contemporaneously — the VAR coefficient links yesterday to today, not today to today. By period 1, Gold has fallen -0.30 from the transmitted shock. This negative effect slowly decays over 4-5 periods.

DXY retains 50% of its shock in period 1, then 22%, then 8%. The momentum halves roughly each period.

This is a complete, calculated IRF. No black box. Just matrix multiplication.

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**[SLIDE 43]** *Visual: 3×3 grid of all 9 IRF combinations.*

In a 3-variable VAR, we have 9 possible impulse response combinations.

Shock Gold → Gold, DXY, S&P each respond. Shock DXY → Gold, DXY, S&P each respond. Shock S&P → Gold, DXY, S&P each respond.

Nine stories. Each revealing a different piece of the market's wiring.

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**[SLIDE 44]** *Visual: Four IRF shape patterns with interpretation labels.*

|                                |                                   |
|--------------------------------|-----------------------------------|
| Quick spike, returns to zero   | → Short-lived shock               |
| Slow build, gradual decay      | → Persistent shock                |
| Initial drop, then recovery    | → Reversal/overshoot effect       |
| Permanent shift, never returns | → ⚠ May indicate non-stationarity |

Learn to read the shapes. If the response NEVER returns to zero, that is a warning — the variables may be cointegrated, which we cover very soon.

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## CHAPTER 7 — ORTHOGONALIZED IRF AND CHOLESKY DECOMPOSITION

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**[SLIDE 45]** *Visual: Sigma matrix with off-diagonal elements highlighted in red. Warning: "Errors are correlated. This is a problem."*

We have a problem.

When we shock DXY in our standard IRF, we are imagining: "DXY gets a shock while Gold stays perfectly calm." But we just proved that is not realistic — their errors have a -0.38 correlation. A real DXY shock arrives with a correlated Gold shock on the same day.

So our standard IRF is telling a slightly wrong story. It pretends shocks happen in isolation when they actually travel together.

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[SLIDE 46] *Visual: Standard IRF vs Orthogonalized IRF — concept comparison.*

The solution: Orthogonalized IRF.

Transform our correlated errors into a new set of errors that ARE uncorrelated with each other. Then we can safely shock one without worrying about the others.

The tool for this transformation is Cholesky Decomposition.

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[SLIDE 47] *Visual: Cholesky decomposition:  $\Sigma = P \cdot P'$ . Sigma shown being factored into lower triangular P and its transpose.*

Cholesky says: any symmetric positive definite matrix can be written as the product of a lower triangular matrix P and its transpose P-prime.

$$\Sigma = P \times P'$$

Where P is lower triangular:

$$P = \begin{bmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{bmatrix}$$

The zero in the upper right is NOT optional. It is the key structural restriction. Let us understand exactly why.

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[SLIDE 48] [NEW] *Visual: Why is the zero there? Left panel: no zero — 4 unknowns, 3 equations. Right panel: zero imposed — 3 unknowns, 3 equations.*

**The zero forces uniqueness.**

Without the zero, P is a full 2×2 matrix: a, b, c, d — four unknowns.

$P \times P'$  gives three equations (because Sigma is symmetric). Four unknowns, three equations — infinite solutions. Every solution implies a completely different story about how Gold and DXY interact. The model would be meaningless.

By setting  $b = 0$ , we have three unknowns: a, c, d. Now exactly three equations. Exactly one solution.

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**[SLIDE 49] [NEW]** Visual: Step-by-step Cholesky calculation with our Sigma =  $\begin{bmatrix} 0.04 & -0.012 \\ -0.012 & 0.025 \end{bmatrix}$ .

**Calculate P for our Sigma:**

$$\text{Sigma} = \begin{vmatrix} 0.040 & -0.012 \\ -0.012 & 0.025 \end{vmatrix}$$

$$p_{11} = \sqrt{0.040} = 0.200$$

$$p_{21} = \text{Sigma}[1,0] / p_{11} = -0.012 / 0.200 = -0.060$$

$$\begin{aligned} p_{22} &= \sqrt{\text{Sigma}[1,1] - p_{21}^2} = \sqrt{0.025 - (-0.060)^2} \\ &= \sqrt{0.025 - 0.0036} = \sqrt{0.0214} = 0.146 \end{aligned}$$

$$P = \begin{vmatrix} 0.200 & 0 \\ -0.060 & 0.146 \end{vmatrix}$$

Verify:  $P \times P' = \begin{vmatrix} 0.200^2 & 0.200 \times (-0.060) \\ (-0.060) \times 0.200 & (-0.060)^2 + 0.146^2 \end{vmatrix} = \begin{vmatrix} 0.040 & -0.012 \\ -0.012 & 0.025 \end{vmatrix}$   
✓ Matches Sigma exactly.

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**[SLIDE 50] [NEW]** Visual: What the zero means economically. And what happens if you use a different ordering.

**What does the zero mean economically?**

$P[0,1] = 0$  means: DXY's structural shock has zero contemporaneous effect on Gold.

We ordered Gold first. Gold moves first within the period. DXY then reacts to Gold, but Gold does not simultaneously react to DXY within the same time step.

This is an assumption grounded in economic reasoning — not a mathematical trick.

**What if we flip the ordering?**

$$P_{\text{flipped}} = \begin{vmatrix} 0.158 & 0.126 \\ -0.075 & 0.131 \end{vmatrix} \leftarrow \text{DXY first}$$

Completely different P matrix. Completely different structural shocks. Completely different IRF results.

Same data. Same Sigma. Different ordering. Different story.

This is the warning: the results depend entirely on your assumed causal ordering. If you have strong economic theory — use it. If you are unsure — use GIRF instead. Coming next.

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[SLIDE 51] [NEW] *Visual: What happens with NO zero — three valid solutions shown side by side, each giving a different IRF.*

Here are three P matrices that all satisfy  $P \times P' = \text{Sigma}$ . All mathematically valid:

Solution 1:  $P = \begin{vmatrix} 0.200 & 0 \\ -0.060 & 0.146 \end{vmatrix}$  (Gold first – Cholesky)

Solution 2:  $P = \begin{vmatrix} 0.158 & 0.126 \\ -0.075 & 0.131 \end{vmatrix}$  (DXY first – Cholesky reversed)

Solution 3:  $P = \begin{vmatrix} -0.180 & 0.088 \\ 0.070 & 0.141 \end{vmatrix}$  (arbitrary rotation)

Each implies a completely different structural interpretation. The data alone cannot tell you which is correct. Economic reasoning must supply the missing constraint — that is the identification problem.

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## CHAPTER 8 — ORTHOGONALIZED IRF: FULL WORKED EXAMPLE [NEW]

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[SLIDE 52] [NEW] *Visual: Theta formula:  $\Theta_h = \Psi_h \times P$ . The meaning of each element.*

The Orthogonalized IRF formula:  $\Theta_h = \Psi_h \times P$

Element  $\Theta_h[i, j]$  = "Response of variable i at horizon h to a one-standard-deviation STRUCTURAL shock to variable j."

The key difference from raw IRF: we are now shocking STRUCTURAL shocks — which are orthogonal (independent) by construction — not correlated reduced-form errors.

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[SLIDE 53] [NEW] *Visual: Theta\_0 calculation.*

$$\Theta_0 = \Psi_0 \times P = I \times P = P$$

$$\Theta_0 = \begin{vmatrix} 0.200 & 0 \\ -0.060 & 0.146 \end{vmatrix}$$

Reading  $\Theta_0$ :

- $\Theta_0[0,0] = 0.200$ : Gold responds to its own structural shock by 0.200% immediately. One standard deviation of Gold's unexplained variation.
- $\Theta_0[0,1] = 0.000$ : Gold has ZERO response to DXY's structural shock at  $h=0$ . This is the zero we imposed. Gold moves first; DXY has not transmitted back yet.

- $\Theta_0[1,0] = -0.060$ : DXY responds to Gold's shock immediately by -0.060%. When Gold jumps, DXY falls slightly within the same period.
- $\Theta_0[1,1] = 0.146$ : DXY responds to its own structural shock by 0.146%.

[SLIDE 54] [NEW] Visual: *Theta\_1* — full matrix multiplication written out.

$$\Theta_1 = \Psi_1 \times P = A \times P$$

$$\theta_1 = \begin{vmatrix} 0.4 & -0.3 \\ 0.1 & 0.5 \end{vmatrix} \times \begin{vmatrix} 0.200 & 0 \\ -0.060 & 0.146 \end{vmatrix}$$

$$\theta_1[0,0] = 0.4 \times 0.200 + (-0.3) \times (-0.060) = 0.080 + 0.018 = 0.098$$

$$\theta_1[0,1] = 0.4 \times 0 + (-0.3) \times 0.146 = 0 - 0.044 = -0.044$$

$$\theta_1[1,0] = 0.1 \times 0.200 + 0.5 \times (-0.060) = 0.020 - 0.030 = -0.010$$

$$\theta_1[1,1] = 0.1 \times 0 + 0.5 \times 0.146 = 0 + 0.073 = 0.073$$

$$\theta_1 = \begin{vmatrix} 0.098 & -0.044 \\ -0.010 & 0.073 \end{vmatrix}$$

$\theta_1[0,1] = -0.044$ : One period after a DXY structural shock, Gold has fallen 0.044%. The transmission took one period to materialize through the VAR dynamics.

[SLIDE 55] [NEW] Visual: *Theta\_2* and *Theta\_3* calculations.

$$\Theta_2 = A^2 \times P \quad (A^2 = [[0.13, -0.27], [0.09, 0.22]])$$

$$\theta_2[0,0] = 0.13 \times 0.200 + (-0.27) \times (-0.060) = 0.026 + 0.016 = 0.042$$

$$\theta_2[0,1] = 0.13 \times 0 + (-0.27) \times 0.146 = 0 - 0.039 = -0.039$$

$$\theta_2[1,0] = 0.09 \times 0.200 + 0.22 \times (-0.060) = 0.018 - 0.013 = 0.005$$

$$\theta_2[1,1] = 0.09 \times 0 + 0.22 \times 0.146 = 0 + 0.032 = 0.032$$

$$\Theta_3 = A^3 \times P \quad (A^3[0,1] = -0.174, A^3[1,0] = 0.058)$$

$$\theta_3[0,1] = (-0.174) \times 0.146 = -0.025$$

$$\theta_3[1,0] = 0.058 \times 0.200 + 0.083 \times (-0.060) = 0.012 - 0.005 = 0.007$$

[SLIDE 56] [NEW] Visual: Complete OIRF summary table.

Complete Orthogonalized IRF Table:

| h | Gold→Gold | DXY→Gold | Gold→DXY | DXY→DXY |
|---|-----------|----------|----------|---------|
| 0 | 0.200     | 0.000    | -0.060   | 0.146   |
| 1 | 0.098     | -0.044   | -0.010   | 0.073   |
| 2 | 0.042     | -0.039   | 0.005    | 0.032   |
| 3 | 0.015     | -0.025   | 0.007    | 0.012   |
| 4 | 0.004     | -0.014   | 0.005    | 0.004   |

Gold's response to its own shock:  $0.200 \rightarrow 0.098 \rightarrow 0.042 \rightarrow 0.015 \rightarrow 0.004$ . Decaying cleanly. By period 4, almost fully dissipated.

Gold's response to DXY shock:  $0.000 \rightarrow -0.044 \rightarrow -0.039 \rightarrow -0.025 \rightarrow -0.014$ . No effect at period 0 (ordering restriction). Then turns negative and stays negative, slowly decaying. A DXY surprise positive shock keeps Gold falling for several periods.

These are the actual impulse response paths. Built from first principles. Just matrix multiplication.

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## CHAPTER 9 — GIRF: THE ORDER-FREE SOLUTION

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[SLIDE 57] *Visual: GIRF title. "Generalized Impulse Response Function — Koop, Pesaran & Potter (1996)."*

What if you do not want to impose a causal ordering? What if you want IRF results completely robust to how you arrange the variables?

In 1996, Koop, Pesaran, and Potter published a paper that solved this problem. They called it the Generalized Impulse Response Function — GIRF.

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[SLIDE 58] *Visual: GIRF formula — conditional expectations.*

$$\text{GIRF}(h, \delta, \Omega_{t-1}) = E[Y_{t+h} \mid \varepsilon_{\text{DXY},t} = \delta, \Omega_{t-1}] - E[Y_{t+h} \mid \Omega_{t-1}]$$

Instead of orthogonalizing errors first, GIRF asks: given that DXY received a specific shock of size  $\delta$ , and given what we already know about the system's history — how much does our forecast of Gold at horizon  $h$  change?

No ordering. No triangular decomposition. No assumptions about who moves first. GIRF uses the historical correlation structure directly, as it is.

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[SLIDE 59] *Visual: OIRF vs GIRF comparison table.*

|                     | OIRF                   | GIRF                 |
|---------------------|------------------------|----------------------|
| Requires ordering?  | Yes                    | No                   |
| Sensitive to order? | Yes (strongly)         | No                   |
| Best for            | Theory-driven analysis | Exploratory analysis |

Use OIRF when you have strong economic theory for your ordering. Use GIRF when you want order-independent, robust results.

OIRF is the detective who follows strict procedure. GIRF is the detective who throws out the rulebook but still solves the case. Both have their place.

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## CHAPTER 10 — LEAD-LAG DISCOVERY

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[SLIDE 60] *Visual: Two zigzag lines — DXY and Gold — one slightly ahead of the other.*

Now for a more direct, practical tool for one specific question: how many days ahead does one variable tend to move before another follows?

The Cross-Correlation Function — CCF — answers this. And it matters enormously for trading.

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[SLIDE 61] *Visual: CCF chart. X-axis: lags -10 to +10. Y-axis: correlation coefficient. Dashed lines at 95% confidence.*

For any two variables X and Y, the CCF at lag k measures: how correlated is X today with Y from k periods ago?

If CCF is highest at lag  $k = +3$ : X leads Y by three days. If highest at lag  $k = -2$ : Y leads X by two days.

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[SLIDE 62] *Visual: CCF result for DXY and Gold — peak at lag +2, negative correlation.*

Reading our CCF between DXY and Gold:

Largest correlation at lag +2, and it is negative.

DXY today is negatively correlated with Gold two days later. When the Dollar rises, Gold tends to fall about two days later.

Dollar leads. Gold follows with a two-day delay.

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[SLIDE 63] *Visual: The result as a simple trading signal.*

This is directly tradable information. If Dollar surges today, a systematic strategy could say: short Gold on day 2.

Not because we are certain. Because statistically, the pattern exists in the data.

The detective is not just solving the case anymore. The detective is also using the case to predict the next crime.

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## CHAPTER 11 — FEVD: FORECAST ERROR VARIANCE DECOMPOSITION

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[SLIDE 64] *Visual: Pie chart — Gold's forecast error: Gold own 65% / DXY 25% / S&P 10%.*

We have seen WHO leads. We have seen HOW the system responds. Now the third big question: who is RESPONSIBLE for what?

Out of all the uncertainty in predicting Gold's future returns — what fraction comes from Gold's own random shocks, what from Dollar shocks, what from S&P shocks?

That is what FEVD answers.

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[SLIDE 65] *Visual: FEVD formula.*

$$\text{FEVD}(i, j, H) = [\sum_{h=0}^{H-1} \theta_h[i, j]^2] / [\sum_j \sum_{h=0}^{H-1} \theta_h[i, j]^2]$$

Numerator: Cumulative squared OIRF of variable  $i$  to shock  $j$

Denominator: Total cumulative squared OIRF of variable  $i$  across ALL shocks

In plain language: "What fraction of Gold's total forecast error, accumulated over  $H$  periods, came from shock  $j$ ?"

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[SLIDE 66] [NEW] *Visual: FEVD at  $h=1$  — fully worked calculation.*

**FEVD for Gold at  $h=1$ :**

At  $h=1$ , only  $\theta_0$  contributes:

From Gold's own shock:  $\theta_0[0,0]^2 = 0.200^2 = \mathbf{0.0400}$  From DXY shock:  $\theta_0[0,1]^2 = 0.000^2 = \mathbf{0.0000}$

Total = 0.0400

FEVD(Gold, own, 1) =  $0.0400 / 0.0400 = \mathbf{100.0\%}$  FEVD(Gold, DXY, 1) =  $0.0000 / 0.0400 = \mathbf{0.0\%}$

At a one-period horizon, ALL of Gold's forecast uncertainty comes from its own shock.

This makes sense. We ordered Gold first — at  $h=0$ , DXY's Theta was zero by construction. So at the very next step, only Gold's own surprise matters.

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[SLIDE 67] [NEW] *Visual: FEVD at  $h=2$  — cumulative, showing DXY contribution appearing.*

### FEVD for Gold at $h=2$ :

Now  $\Theta_0$  AND  $\Theta_1$  both contribute:

From Gold's own shock:  $\Theta_0[0,0]^2 + \Theta_1[0,0]^2 = 0.200^2 + 0.098^2 = 0.0400 + 0.0096 = \mathbf{0.0496}$

From DXY shock:  $\Theta_0[0,1]^2 + \Theta_1[0,1]^2 = 0.000^2 + (-0.044)^2 = 0.0000 + 0.0019 = \mathbf{0.0019}$

Total =  $0.0496 + 0.0019 = 0.0515$

FEVD(Gold, own, 2) =  $0.0496 / 0.0515 = \mathbf{96.3\%}$  FEVD(Gold, DXY, 2) =  $0.0019 / 0.0515 = \mathbf{3.7\%}$

Notice what happened. As soon as we look one more period ahead, DXY's contribution appeared — 3.7%. Because the DXY shock from period 0 now had one period to transmit through the VAR coefficients into Gold's forecast error.

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[SLIDE 68] [NEW] *Visual: Full FEVD table built from calculations.*

#### FEVD for GOLD:

| H   | Own Shock | DXY Shock | What's happening                    |
|-----|-----------|-----------|-------------------------------------|
| --- | -----     | -----     | -----                               |
| 1   | 100.00%   | 0.00%     | Only $\Theta_0$ — DXY zero at $h=0$ |
| 2   | 96.26%    | 3.74%     | $\Theta_1[0,1]^2=0.044^2$ adds DXY  |
| 3   | 93.65%    | 6.35%     | $\Theta_2[0,1]^2=0.039^2$ adds more |
| 5   | 92.27%    | 7.73%     | DXY influence grows and stabilizes  |
| 8   | 92.19%    | 7.81%     | Near convergence                    |

#### FEVD for DXY:

| H   | Gold Shock | Own Shock |                                                                       |
|-----|------------|-----------|-----------------------------------------------------------------------|
| --- | -----      | -----     |                                                                       |
| 1   | 14.40%     | 85.60%    | ← Gold hits DXY contemporaneously<br>( $\theta_0[1,0]=-0.060\neq 0$ ) |
| 5   | 11.94%     | 88.06%    |                                                                       |

Gold is mostly its own story — 92% even at long horizons. DXY contributes a stable 7-8% as we look further ahead.

DXY is 88% own-shock, but Gold contributes 12-14% immediately — because we ordered Gold first, Gold hits DXY contemporaneously.

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**[SLIDE 69]** *Visual: Stacked bar chart of FEVD across horizons — each bar 100%, colored by variable contribution.*

At short horizons: Gold is dominated by its own noise. At long horizons: Dollar becomes a growing source of uncertainty.

This tells us: short-term Gold trader? Focus on Gold's own momentum. Long-term Gold investor? Watch the Dollar closely.

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**[SLIDE 70]** **[NEW]** *Visual: FEVD as a trade sizing tool. Two traders with different position sizes.*

### **How does FEVD actually improve trade calls?**

Imagine you are a Gold options trader. You want to buy a straddle — a bet that Gold will move a lot. You need to size it.

**Without FEVD:** You look at Gold's historical volatility:  $\pm 0.20\%$  per day. You size the straddle based on Gold's own volatility alone.

**With FEVD:** You know that at a 5-day horizon, 7.7% of Gold's forecast error variance comes from DXY shocks.

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**[SLIDE 71]** **[NEW]** *Visual: Fed meeting scenario — DXY options pricing implies high Dollar uncertainty.*

Now add this: there is a Federal Reserve meeting in three days. DXY options are very expensive — the market is pricing a possible 2-3% Dollar move in either direction.

Without FEVD: you price your Gold straddle using Gold's own historical volatility only. Small straddle.

With FEVD: you know 7.7% of Gold's 5-day forecast variance is driven by DXY shocks. If DXY's variance is about to spike threefold, Gold's total forecast variance at the 5-day horizon increases by approximately  $0.077 \times 3 = \mathbf{23\% \text{ above normal}}$ .

You buy a larger straddle. You are compensated for the macro risk that FEVD revealed.

FEVD does not tell you which direction Gold moves. It tells you how much of Gold's uncertainty is borrowed from other markets. When those other markets get volatile — due to policy events, geopolitical shocks, supply disruptions — Gold is dragged along in proportion to the FEVD weight.

That proportion is directly actionable for position sizing.

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[SLIDE 72] *Visual: IRF vs FEVD side by side — path vs scorecard.*

FEVD is a cross-market sensitivity dashboard. It tells you where to look when sizing positions.

IRF = the story: "How does Gold respond to a Dollar shock over time?" FEVD = the scorecard: "Over all shocks combined, how much does Dollar matter to Gold's uncertainty?"

Two sides of the same coin. IRF is the story. FEVD is the attribution.

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## CHAPTER 12 — COVARIANCE VS CORRELATION: THE SETUP

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[SLIDE 73] *Visual: COVARIANCE vs CORRELATION — two boxes with "vs" between them.*

Before cointegration, let us clear up something that confuses people constantly.

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[SLIDE 74] *Visual: Two scatter plots — absolute price levels vs returns scaled.*

Covariance measures how two variables move together. Positive means they move in the same direction. Negative means opposite.

The problem: covariance is in the original units of the data. A Gold-DXY covariance number in raw prices is uninterpretable on its own.

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[SLIDE 75] *Visual: Correlation formula = Covariance / ( $\sigma_X \times \sigma_Y$ ). Always between -1 and +1.*

Correlation fixes this — divides by both standard deviations. Always between -1 and +1. Now you can compare across any pair of variables regardless of units.

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[SLIDE 76] *Visual: Critical insight box — both measure short-run co-movement only.*

Here is the critical point — your setup for cointegration.

Both covariance and correlation only measure **short-run co-movement**. They tell you: on a day-to-day basis, do these two variables tend to move together?

They do NOT tell you: is there a long-run relationship that keeps these variables bound together even when they wander apart in the short run?

That long-run bond is cointegration. A completely different concept.

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## CHAPTER 13 — COINTEGRATION: THE LONG-RUN BOND

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[SLIDE 77] *Visual: Two people walking randomly while holding hands — each wanders but their relative distance stays bounded.*

Let me tell you a story.

Two people who have both had too much to drink and are walking home. Each one walking randomly — left, right, forward, backward. Completely unpredictable in the short run. But they are holding hands.

Even though each person's path looks like random noise on its own, their relative distance to each other stays bounded. When they drift apart, the connection pulls them back.

**That connection is cointegration.** Two non-stationary series that wander randomly on their own, but share a long-run bond that prevents them from drifting apart forever.

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[SLIDE 78] *Visual: Gold and Silver prices over 8 days — both trending. The spread bouncing around zero.*

The classic example: Gold and Silver.

Both trend upward. Neither is stationary on its own.

But the spread — Gold minus 60.5  $\times$  Silver — stays roughly mean-reverting. It wanders but always comes back.

That spread is stationary. The existence of a stationary linear combination is exactly what cointegration means.

---

[SLIDE 79] [NEW] *Visual: OLS regression of Gold on Silver — full calculation of beta.*

**Where does the coefficient 60.5 come from?**

It is NOT chosen randomly. It is calculated via OLS regression:  $\text{Gold} = \alpha + \beta \times \text{Silver}$ .

$$\beta = \text{Cov}(\text{Gold}, \text{Silver}) / \text{Var}(\text{Silver})$$

Using 8 data points (Gold: 1800-1860, Silver: 30.0-31.0):

Products (Gold<sub>i</sub> - 1817.5)(Silver<sub>i</sub> - 30.26):

Day 1: (-17.5)×(-0.26) = +4.55

Day 2: (+2.5)×(+0.04) = +0.10

Day 3: (-27.5)×(-0.46) = +12.65

Day 4: (+32.5)×(+0.54) = +17.55

Day 5: (-7.5)×(-0.16) = +1.20

Day 6: (+22.5)×(+0.34) = +7.65

Day 7: (-47.5)×(-0.76) = +36.10

Day 8: (+42.5)×(+0.74) = +31.45

Sum = 111.25

$$\text{Cov}(\text{Gold}, \text{Silver}) = 111.25 / 7 = 15.89$$

$$\text{Var}(\text{Silver}) = 0.2626$$

$$\beta = 15.89 / 0.2626 = 60.5 \leftarrow \text{Calculated from data, not assumed.}$$

$$\alpha = 1817.5 - 60.5 \times 30.26 = -13.2$$

The cointegrating coefficient  $\beta = 60.5$  comes from the data. It is the slope that makes the linear combination stationary.

**[SLIDE 80] [NEW]** *Visual: The spread computed for each day. Plot showing it oscillating around zero.*

$$\text{Spread} = \text{Gold} - 60.5 \times \text{Silver} - (-13.2) = \text{Gold} - 60.5 \times \text{Silver} + 13.2$$

| Day | Gold | Silver | Spread |
|-----|------|--------|--------|
| 1   | 1800 | 30.0   | -1.62  |
| 2   | 1820 | 30.3   | +0.23  |
| 3   | 1790 | 29.8   | +0.48  |
| 4   | 1850 | 30.8   | -0.02  |
| 5   | 1810 | 30.1   | +2.33  |
| 6   | 1840 | 30.6   | +2.08  |
| 7   | 1770 | 29.5   | -1.37  |
| 8   | 1860 | 31.0   | -2.12  |

Spread bounces between -2.12 and +2.33. No trend. Always reverting toward zero.

This is **mean reversion** — whenever the spread wanders away from zero, it tends to come BACK toward zero. The spread has memory of where it belongs.

## Trading application:

Spread +2.50 today (Gold expensive relative to Silver) → SELL Gold, BUY Silver. Exit when spread returns near zero.

Spread -3.00 today (Gold cheap relative to Silver) → BUY Gold, SELL Silver. Exit when spread returns near zero.

Stationarity of the spread is what makes the trade viable. If the spread were non-stationary — wandering forever — it might never come back, and the trade would never close.

---

[SLIDE 81] *Visual: Spurious regression warning — two completely unrelated trended series with  $R^2 = 0.87$ .*

Big warning. If you run a regular regression on two non-stationary series — even completely unrelated ones — you often get high R-squared and significant coefficients.

This is called spurious regression. Granger and Newbold showed in 1974 that regressing two independent random walks on each other produces significant t-statistics about 75% of the time.

75% false alarms.

Cointegration testing is how we distinguish real long-run relationships from coincidental shared trends.

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[SLIDE 82] *Visual: Engle-Granger test — 2-step procedure.*

Two main cointegration tests:

**Engle-Granger** — simple, limited to two variables. Step 1: regress Y on X, save residuals. Step 2: test if those residuals are stationary using ADF. Stationary residuals → cointegration confirmed.

Good for pairs. Not suitable for more than two variables.

---

[SLIDE 83] [NEW] *Visual: Johansen test procedure as 5 steps.*

**Johansen test** — multivariate, handles 3+ variables, gives you the cointegration rank. Here is the actual procedure:

- Step 1: Run two auxiliary regressions to get residuals  $R_0$  and  $R_1$
  - Step 2: Compute moment matrices  $S_{00}$ ,  $S_{01}$ ,  $S_{11}$  from those residuals
  - Step 3: Solve the eigenvalue problem  $|\lambda S_{11} - S_{10} S_{00}^{-1} S_{01}| = 0$
  - Step 4: Order eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$
  - Step 5: Compute Trace statistics and compare to critical values

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[SLIDE 84] [NEW] *Visual: Step 1 in detail — what  $R_0$  and  $R_1$  are.*

**Step 1: Two auxiliary regressions.**

**Regression A:** Regress  $\Delta Y_t$  (first differences — the returns) on lagged differences  $\Delta Y_{t-1}$ ,  $\Delta Y_{t-2}$ . Save residuals  $\rightarrow R_0t$  — "the change in Y today, after removing short-run VAR dynamics."

**Regression B:** Regress  $Y_{t-1}$  (the LEVELS, lagged one period) on those same lagged differences. Save residuals  $\rightarrow R_1t$  — "the lagged level of Y, after removing short-run dynamics."

By regressing out the short-run dynamics, we isolate the pure long-run signal.  $R_0$  and  $R_1$  are the cleaned, de-correlated versions of changes and levels.

---

[SLIDE 85] [NEW] *Visual: Step 2 — moment matrices.  $S_{00}$ ,  $S_{01}$ ,  $S_{11}$  formulas.*

**Step 2: Moment matrices.**

$$\begin{aligned} S_{00} &= (1/T) \times \sum R_0t \times R_0t' && \text{(variance of the change residuals)} \\ S_{01} &= (1/T) \times \sum R_1t \times R_0t' && \text{(covariance between changes and levels)} \\ S_{11} &= (1/T) \times \sum R_1t \times R_1t' && \text{(variance of the level residuals)} \end{aligned}$$

If  $R_0$  and  $R_1$  are highly related — if changes in Y are strongly predicted by the level of Y — that is the signature of cointegration. The eigenvalue problem extracts exactly how strong that signal is.

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[SLIDE 86] [NEW] *Visual: Step 5 — Trace statistics fully calculated with eigenvalues [0.058, 0.022, 0.006],  $T=500$ .*

**Step 5: Trace statistics — fully worked.**

Our system:  $T = 500$  observations. Eigenvalues from the procedure:

$$\lambda_1 = 0.058 \quad \lambda_2 = 0.022 \quad \lambda_3 = 0.006$$

**Test  $H_0$ : rank  $\leq 0$  (no cointegration):**

$$\begin{aligned}
\text{Trace}(0) &= -T \times [\ln(1-\lambda_1) + \ln(1-\lambda_2) + \ln(1-\lambda_3)] \\
&= -500 \times [\ln(0.942) + \ln(0.978) + \ln(0.994)] \\
&= -500 \times [-0.05981 + (-0.02224) + (-0.00602)] \\
&= -500 \times (-0.08807) = 44.03
\end{aligned}$$

Critical value (5%):  $29.80 \rightarrow 44.03 > 29.80 \rightarrow \text{REJECT}$  

At least one cointegrating relationship exists.

**Test  $H_0$ : rank  $\leq 1$ :**

$$\begin{aligned}
\text{Trace}(1) &= -500 \times [\ln(0.978) + \ln(0.994)] \\
&= -500 \times [-0.02224 + (-0.00602)] \\
&= -500 \times (-0.02826) = 14.13
\end{aligned}$$

Critical value:  $15.49 \rightarrow 14.13 < 15.49 \rightarrow \text{FAIL TO REJECT}$  

**Conclusion: Cointegration rank = 1. Exactly one long-run relationship.**

The eigenvector corresponding to  $\lambda_1 = 0.058$  IS the cointegrating vector  $\beta$ . That is the combination you use in your VECM.

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## CHAPTER 14 — VECM: CORRECTING THE DRIFT

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**[SLIDE 87]** *Visual: The drunk couple from earlier — now with a leash connecting them.*

We have confirmed cointegration. There is a long-run bond.

But the variables still wander in the short run. What brings them back?

The Vector Error Correction Model — VECM — adds an explicit correction mechanism to the VAR framework.

Every period, if the variables have drifted away from their long-run relationship, each variable adjusts — at its own speed — to pull the system back toward equilibrium.

---

**[SLIDE 88] [NEW]** *Visual: VECM equation fully annotated — every term labeled.*

Before reading the equation, one important clarification.

**What is  $\Delta Y_t$ ?**

If  $Y_t = \log(\text{Price}_t)$ , then:

$$\Delta Y_t = \log(\text{Price}_t) - \log(\text{Price}_{t-1}) = \log(\text{Price}_t / \text{Price}_{t-1}) = \text{the log return.}$$

So  $\Delta Y_t$  IS the log return. It is stationary.  $\Delta Y_{t-1}$  is yesterday's log return. Also stationary.

But  $Y_{t-1}$  (without the delta) is the log price LEVEL. Non-stationary.

VECM mixes both: log returns on the left side AND log levels inside the error correction term.

---

[SLIDE 89] [NEW] Visual: Full VECM equation for 2-variable system. Every term labeled with color coding.

$$\begin{aligned}\Delta \text{Gold}_t &= c_1 + \Gamma_{11} \cdot \Delta \text{Gold}_{t-1} + \Gamma_{12} \cdot \Delta \text{Silver}_{t-1} \\ &\quad + \alpha_1 \cdot (\beta_1 \cdot \text{Gold}_{t-1} + \beta_2 \cdot \text{Silver}_{t-1} - \mu) + \varepsilon_{1t} \\ \Delta \text{Silver}_t &= c_2 + \Gamma_{21} \cdot \Delta \text{Gold}_{t-1} + \Gamma_{22} \cdot \Delta \text{Silver}_{t-1} \\ &\quad + \alpha_2 \cdot (\beta_1 \cdot \text{Gold}_{t-1} + \beta_2 \cdot \text{Silver}_{t-1} - \mu) + \varepsilon_{2t}\end{aligned}$$

Every piece decoded:

$\Delta \text{Gold}_t$  — Gold's LOG RETURN today → stationary ✓

$\Gamma_{11} \cdot \Delta \text{Gold}_{t-1}$  — yesterday's Gold return × coefficient → short-run momentum

$\Gamma_{12} \cdot \Delta \text{Silver}_{t-1}$  — yesterday's Silver return → short-run cross-market spillover

$\alpha_1$  — Gold's speed of adjustment → how fast Gold corrects when system is out of equilibrium

$(\beta_1 \cdot \text{Gold}_{t-1} + \beta_2 \cdot \text{Silver}_{t-1} - \mu)$  — **THE ERROR CORRECTION TERM** → how far are the LOG LEVELS from their long-run relationship right now? This is the spread from yesterday's level values.

$\varepsilon_{1t}$  — unexplained surprise today

The error correction term uses  $\text{Gold}_{t-1}$  and  $\text{Silver}_{t-1}$  — log LEVELS. Yesterday's levels reveal whether the system is above or below its long-run equilibrium.

---

[SLIDE 90] [NEW] Visual: Numerical example — one specific day plugged into the VECM.

Let us plug in numbers for one day.

Suppose yesterday:  $\text{ECT}_{t-1} = \text{Gold}_{t-1} - 60.5 \times \text{Silver}_{t-1} + 13.2 = +0.5$  (above equilibrium)

Estimated adjustment speed:  $\alpha_1 = -0.12$

Correction contribution to Gold's return today:  $\alpha_1 \times \text{ECT}_{t-1} = -0.12 \times 0.5 = -0.06$

Gold's return today is pulled DOWN by 0.06% because Gold was 0.5 units too high relative to Silver yesterday.

This is the error correction mechanism — a mechanical pull toward equilibrium. Not mystical. Not assumed. Just multiplication.

---

[SLIDE 91] [NEW] *Visual: How  $\beta$  (the cointegrating vector) is calculated — from the Johansen eigenvector.*

### How is $\beta$ actually calculated?

$\beta$  is the eigenvector corresponding to the LARGEST eigenvalue  $\lambda_1$  from the Johansen procedure. It is not imposed by the researcher — it is estimated from the data.

For a simple 2-variable case, the eigenvector  $v$  satisfies:

$$(S_{10} \cdot S_{00}^{-1} \cdot S_{01}) \times v = \lambda_1 \times S_{11} \times v$$

The software solves this system and returns  $v$ . We normalize so the first element equals 1 — this gives  $\beta = [1, -60.5]$  in our case.

The coefficient -60.5 is what makes Gold - 60.5×Silver stationary. The data chose it. Not us.

---

[SLIDE 92] *Visual: Alpha vector — adjustment speeds.*

$\alpha_{\text{Gold}} = -0.12 \quad \rightarrow \text{Gold adjusts -12\% per period toward equilibrium}$   
 $\alpha_{\text{Silver}} = +0.04 \quad \rightarrow \text{Silver adjusts +4\% per period}$

The SIGN of alpha tells you the direction of correction. The MAGNITUDE tells you the speed. Larger magnitude = faster mean reversion = faster spread trading signals.

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[SLIDE 93] *Visual: Rubber band analogy — stronger rubber band = faster correction.*

Think of alpha as the strength of the rubber band connecting variables to their long-run path. Strong rubber band (large alpha) means deviations are quickly corrected. Weak rubber band means deviations can persist for a long time.

---

[SLIDE 94] [NEW] *Visual: Decision tree — expanded with explicit guidance about when to use VECM if returns are stationary.*

Are variables stationary (returns/differences)?

↓ YES

Use VAR in returns. This is valid and fast.

Are the LEVELS cointegrated?

↓ YES

VAR in returns is valid – but loses long-run equilibrium information.

VECM captures BOTH short-run AND long-run. Preferred.

↓ NO

Use VAR in returns. Correct.

Are variables non-stationary AND non-cointegrated?

→ Difference all series. Use VAR in differences.

Are variables non-stationary AND cointegrated?

→ Use VECM. Do not difference – you would lose cointegration information.

**The common question:** "If returns are already stationary, why ever use VECM?"

**Answer:** VAR in returns does not know about the current level of the spread. If Gold and Silver are 5 standard deviations apart from their long-run equilibrium, VAR in returns is completely blind to this. VECM explicitly measures the disequilibrium and feeds it into tomorrow's expected return. VAR in returns trades short-run dynamics. VECM trades the long-run bond AND the short-run dynamics.

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## CHAPTER 15 — SVAR: THE CHAIN OF COMMAND

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[SLIDE 95] *Visual: Interrogation room with three suspects — "who gave the orders?"*

We have estimated VAR. We have impulse responses. We have FEVD. We have the error correction.

But all these tools used the reduced-form VAR — we ran regressions without making strong claims about the causal structure.

Now we want to go deeper: what is the TRUE causal structure? Not just "DXY Granger-causes Gold" — but through what mechanism? What is the chain of command?

This is where Structural VAR — SVAR — enters.

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[SLIDE 96] *Visual: Reduced-form vs Structural VAR — the B matrix.*

In reduced-form VAR:  $Y_t = c + A_1 Y_{t-1} + \varepsilon_t$  (reduced-form errors)

In SVAR: the reduced-form errors are generated by deeper structural shocks  $u_t$  through a matrix  $B$ :

$$\epsilon_t = B \times u_t$$

Where  $u_t$  are the true structural shocks — demand shocks, supply shocks, policy shocks — that are genuinely independent of each other.

---

[SLIDE 97] *Visual: Three structural shocks labeled.*

$u_{1_t}$  = Gold-specific demand shock (safe-haven demand)  
 $u_{2_t}$  = Dollar policy shock (Fed decision)  
 $u_{3_t}$  = Risk appetite shock (risk-on/risk-off)

These three underlying forces generate all the correlation and movement we observe.

---

[SLIDE 98] [NEW] *Visual: The 9-unknown, 6-equation problem shown explicitly.*

**The Identification Problem — shown precisely.**

You know Sigma. You estimated it. It looks like:

Sigma = | 0.040    -0.012    0.008 |  
          | -0.012    0.025    -0.005 |  
          | 0.008    -0.005    0.030 |

3 diagonal + 3 off-diagonal = **6 unique numbers**. That is what the data gives you.

You want to find  $B$  such that  $\text{Sigma} = B \times B'$  (assuming structural shocks are uncorrelated):

$B =$  |  $b_{11}$     $b_{12}$     $b_{13}$  |    $\leftarrow$  3 unknowns in row 1  
      |  $b_{21}$     $b_{22}$     $b_{23}$  |    $\leftarrow$  3 unknowns in row 2  
      |  $b_{31}$     $b_{32}$     $b_{33}$  |    $\leftarrow$  3 unknowns in row 3  
                                    Total: 9 unknowns

$B \times B'$  gives a  $3 \times 3$  symmetric matrix — only 6 unique equations.

**9 unknowns – 6 equations = 3 restrictions needed.**

For any  $n \times n$  SVAR:  $n(n-1)/2$  restrictions needed for exact identification.

For  $n=3$ :  $3 \times 2 / 2 =$  **3 restrictions**. Exactly what we calculated. You must supply 3 economic assumptions to make the model uniquely identified.

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[SLIDE 99] *Visual: Three identification methods listed.*

Three main identification approaches:

1. **Short-run restrictions (Cholesky)** — impose zeros in B within the same time period
  2. **Long-run restrictions (Blanchard-Quah)** — impose zeros on permanent cumulative effects
  3. **Sign restrictions** — impose signs (+/-) on responses rather than exact zeros
- 

[SLIDE 100] *Visual: Short-run restriction — lower triangular B matrix.*

$$\text{Short-run: } B = \begin{bmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$$

Within the same period: shock 1 affects all. Shock 2 affects variables 2 and 3 only. Shock 3 affects only variable 3.

This is equivalent to Cholesky ordering. Three zeros imposed = three restrictions. Exactly identified.

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[SLIDE 101] [NEW] *Visual: Long-run restrictions — Blanchard-Quah approach.*

**Long-run restrictions — the Blanchard-Quah (1989) approach:**

The restriction: certain structural shocks have NO PERMANENT effect on certain variables.

Classic example: demand shocks cannot permanently change the level of real output. Only supply shocks can do that permanently. This is a defensible economic claim.

**How to translate into math:**

The cumulative long-run effect matrix is:

$$C = (I - A_1 - A_2 - \dots)^{-1} \quad [\text{For VAR}(1): C = (I - A)^{-1}]$$

$$\text{For } A = \begin{bmatrix} 0.4 & -0.3 \\ 0.1 & 0.5 \end{bmatrix}:$$

$$I - A = \begin{bmatrix} 0.6 & 0.3 \\ -0.1 & 0.5 \end{bmatrix}$$

$$\det(I-A) = 0.6 \times 0.5 - 0.3 \times (-0.1) = 0.30 + 0.03 = 0.33$$

$$C = (I-A)^{-1} = (1/0.33) \times \begin{bmatrix} 0.5 & -0.3 \\ 0.1 & 0.6 \end{bmatrix} = \begin{bmatrix} 1.515 & -0.909 \\ 0.303 & 1.818 \end{bmatrix}$$

The structural long-run IRF matrix is:  $L = C \times B$

The Blanchard-Quah restriction says:  $L[0, 1] = 0$  (shock 2 — demand — has zero PERMANENT effect on variable 1 — output).

$$(C \times B)[0, 1] = 1.515 \times B[0, 1] + (-0.909) \times B[1, 1] = 0$$
$$\rightarrow B[0, 1] = (0.909 / 1.515) \times B[1, 1] = 0.600 \times B[1, 1]$$

One equation. One restriction. Combined with 6 from Sigma: exactly identified.

### Key difference from Cholesky:

Cholesky: "Variable 2 cannot affect variable 1 INSTANTANEOUSLY." Blanchard-Quah: "Shock 2 cannot affect variable 1 PERMANENTLY."

Much more economically defensible. You are not claiming the Fed announcement had no immediate market impact. You are claiming demand shocks do not permanently change output capacity.

---

[SLIDE 102] *Visual: Sign restrictions table.*

Sign restrictions:

|        | Gold Shock | Dollar Shock | Risk Shock |
|--------|------------|--------------|------------|
| Gold   | +          | -            | +          |
| DXY    | -          | +            | -          |
| S&P500 | -          | -            | +          |

Sign restrictions impose directions rather than exact zeros. Flexible — no exact ordering required. Modern SVAR increasingly uses this approach for exactly that reason.

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[SLIDE 103] *Visual: Structural IRF plots — clean, economically interpretable.*

Once SVAR is identified, we get structural impulse response functions — the truly interpretable IRFs.

Not "how does Gold respond to a mathematical transformation of its correlated errors" — but "how does it respond to a real, isolated monetary policy shock with a real economic interpretation?"

This is the gold standard of VAR analysis. Pun absolutely intended.

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## CHAPTER 16 — STRUCTURAL FEVD

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**[SLIDE 104]** *Visual: Structural FEVD table — columns are structural shocks, not variables.*

Gold's forecast variance explained by:

| Horizon | Safe-Haven Shock | Dollar Policy Shock | Risk Shock |
|---------|------------------|---------------------|------------|
| 1       | 72%              | 18%                 | 10%        |
| 10      | 55%              | 30%                 | 15%        |
| 20      | 48%              | 35%                 | 17%        |

Same concept as FEVD — but now we decompose into STRUCTURAL shocks rather than variable shocks.

Instead of "30% of Gold's uncertainty comes from DXY" — we now say "35% of Gold's long-run uncertainty comes from Dollar POLICY shocks specifically."

One tells you which variable. The other tells you which underlying economic force.

Structural FEVD is what central bank research departments use when they say "most of the movement in this market is driven by monetary policy uncertainty."

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**[SLIDE 105]** *Visual: Checkpoint recap — all tools pinned on detective board.*

VAR  → Granger Causality  → IRF  → OIRF  → GIRF   
→ Lead-Lag  → FEVD  → Cointegration  → VECM   
→ SVAR  → Structural FEVD 

Let us pause and look at how far we have come. Every tool we have built so far assumed one thing: parameters are constant over time.

Now we deal with the final question: what if the relationships change?

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## CHAPTER 17 — TIME-VARYING VAR

**[SLIDE 106]** *Visual: A VAR coefficient plotted over time — drifting, changing, not constant.*

During the 2008 crisis, Gold and the Dollar moved with very different dynamics than during calm 2015. During COVID 2020, correlations changed dramatically. Old relationships broke down. New ones emerged.

A fixed-parameter VAR cannot capture this.

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**[SLIDE 107]** *Visual: Rolling window concept — a window slides over the data, re-estimating at each step.*

The simplest solution: rolling window VAR.

Pick a window — say 250 trading days. Estimate VAR using only the most recent 250 observations. Move forward one day. Re-estimate. Plot how coefficients change over time.

Simple. Transparent. But transition between windows is abrupt.

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**[SLIDE 108]** *Visual: TVP-VAR — coefficients drift smoothly, like a slow gradient, not abrupt jumps.*

More sophisticated: Time-Varying Parameter VAR — TVP-VAR.

Parameters drift gradually over time, following a random walk:  $A_{1,t} = A_{1,t-1} + \eta_t$  (small random drift each period)

Estimated using a Kalman Filter — a technique that updates parameter estimates in real time as new data arrives.

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**[SLIDE 109]** *Visual: Time-varying IRF — same shock, different response curves at three different dates.*

The result: time-varying impulse responses.

Same shock to the Dollar. Very different Gold response in 2008 versus 2015 versus 2020.

In 2008: large and persistent. Fear dominant. In 2015: smaller, quicker to dissipate. In 2020: enormous initially, then reversed as both Dollar and Gold moved together.

Same model structure. Same three variables. Completely different economic regimes.

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## CHAPTER 18 — MARKOV-SWITCHING VAR

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**[SLIDE 110]** *Visual: Two clear regimes — CALM (green) and CRISIS (red). An abrupt switch between them.*

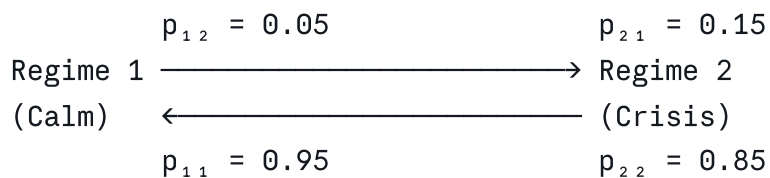
TVP-VAR allows parameters to drift continuously. But sometimes markets do not drift — they SWITCH abruptly.

One day: calm, low volatility, normal correlations. Next week: crisis, high volatility, correlations flip.

The model that captures regime changes is Markov-Switching VAR.

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**[SLIDE 111]** *Visual: Markov chain — two states with transition probabilities.*



If in calm regime today: 95% chance of staying calm tomorrow. If in crisis regime today: 85% chance of staying in crisis.

Crisis states are persistent — once you enter, hard to exit quickly.

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**[SLIDE 112]** *Visual: Two regime-specific VAR equations side by side.*

$$\begin{aligned} \text{Regime 1 (Calm): } Y_t &= c_1 + A_1^{(1)} Y_{t-1} + \varepsilon_t^{(1)} && (\text{small } \sigma_1) \\ \text{Regime 2 (Crisis): } Y_t &= c_2 + A_1^{(2)} Y_{t-1} + \varepsilon_t^{(2)} && (\text{large } \sigma_2) \end{aligned}$$

Completely separate VARs in each regime. Relationships between Gold, Dollar, and S&P 500 are DIFFERENT in each state.

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**[SLIDE 113]** *Visual: K=1 special case — Markov-Switching VAR collapses to standard VAR.*

And here is the beautiful special case: when  $K = 1$  — only one regime — the Markov chain becomes trivial. Nowhere to switch to.

The Markov-Switching VAR collapses to exactly the standard VAR we started with.

One regime. Constant parameters. The model we began with in Chapter 1.

The standard VAR is the simplest member of the Markov-Switching VAR family.

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**[SLIDE 114]** **[NEW]** *Visual: Hamilton Filter — two-step Bayesian update at each time period.*

### **The Hamilton Filter — step by step.**

The Hamilton Filter estimates the probability of being in each regime at each date. It runs two steps every period:

Setup:

- Regime 1 (Calm): mean return +0.05%, daily std 2.0%
  - Regime 2 (Crisis): mean return -0.20%, daily std 5.0%
  - Transition:  $P(\text{stay calm}) = 0.95$ ,  $P(\text{stay crisis}) = 0.85$
  - Starting belief:  $P(\text{Calm}) = 0.80$ ,  $P(\text{Crisis}) = 0.20$
-

[SLIDE 115] [NEW] *Visual: Day 1 — calm +0.1% return. Full predict and update calculation.*

**Day 1: Observed return = +0.1%**

**LIKELIHOOD step** — how likely is +0.1% in each regime?

$$\begin{aligned} f(+0.1\% \mid \text{Calm}) &: z = (0.001 - 0.0005) / 0.02 = 0.025 \rightarrow f = 19.94 \\ f(+0.1\% \mid \text{Crisis}) &: z = (0.001 - (-0.002)) / 0.05 = 0.06 \rightarrow f = 2.09 \end{aligned}$$

A +0.1% return is nearly 10× more likely under the calm regime.

**UPDATE step:**

$$\begin{aligned} \text{Joint: Calm} &= 0.80 \times 19.94 = 15.95 \\ \text{Crisis} &= 0.20 \times 2.09 = 0.418 \\ \text{Total} &= 16.37 \\ \\ \text{Updated: } P(\text{Calm} \mid \text{data}) &= 15.95 / 16.37 = 0.974 \\ P(\text{Crisis} \mid \text{data}) &= 0.418 / 16.37 = 0.026 \end{aligned}$$

After a quiet +0.1% day: 97.4% confident we are in the calm regime.

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[SLIDE 116] [NEW] *Visual: Day 2 — crash -5.0%. Full two-step calculation showing regime probability flip.*

**Day 2: Observed return = -5.0%**

**PREDICT step** (using yesterday's updated probabilities):

$$\begin{aligned} P(\text{Calm today}) &= 0.974 \times 0.95 + 0.026 \times 0.15 = 0.925 + 0.004 = 0.929 \\ P(\text{Crisis today}) &= 0.974 \times 0.05 + 0.026 \times 0.85 = 0.049 + 0.022 = 0.071 \end{aligned}$$

Before seeing today's data, we still expect mostly calm — yesterday was calm.

**LIKELIHOOD step:**

$$\begin{aligned} f(-5\% \mid \text{Calm}) &: z = (-0.05 - 0.0005) / 0.02 = -2.525 \rightarrow f = 0.821 \\ f(-5\% \mid \text{Crisis}) &: z = (-0.05 - (-0.002)) / 0.05 = -0.96 \rightarrow f = 7.963 \end{aligned}$$

A -5% return is nearly 10× more likely under the crisis regime.

**UPDATE step:**

$$\begin{aligned}\text{Joint: Calm} &= 0.929 \times 0.821 = 0.763 \\ \text{Crisis} &= 0.071 \times 7.963 = 0.565 \\ \text{Total} &= 1.328\end{aligned}$$

$$\begin{aligned}\text{Updated: } P(\text{Calm}|\text{data}) &= 0.763/1.328 = 0.574 \\ P(\text{Crisis}|\text{data}) &= 0.565/1.328 = 0.426\end{aligned}$$

One large crash moved us from 97% calm to 57% calm. The model now assigns 43% probability to the crisis regime. Another crash tomorrow would push it above 80%.

This is Bayesian updating. One day at a time. Using the likelihood of the observation under each regime.

[SLIDE 117] [NEW] *Visual: Kalman Filter setup for TVP-VAR — state space equations.*

**The Kalman Filter — for TVP-VAR continuously drifting parameters.**

The Kalman Filter tracks continuously drifting parameters rather than discrete regime switches.

$$\begin{aligned}\text{Observation equation: } Y_t &= X_t \cdot \theta_t + \varepsilon_t && (\text{VAR with time-varying } \theta) \\ \text{State equation: } \theta_t &= \theta_{t-1} + \eta_t && (\text{parameters drift as random walk}) \\ \varepsilon_t &\sim N(0, R) && \leftarrow \text{observation noise} \\ \eta_t &\sim N(0, Q) && \leftarrow \text{parameter drift noise}\end{aligned}$$

At each time  $t$ , two steps:

**PREDICT:** Uncertainty grows because parameters may have drifted.

$$\begin{aligned}\theta_{t|t-1} &= \theta_{t-1|t-1} && (\text{estimate unchanged}) \\ P_{t|t-1} &= P_{t-1|t-1} + Q && (\text{uncertainty increases by } Q)\end{aligned}$$

**UPDATE:** New data revises the estimate.

$$\begin{aligned}\text{Kalman Gain: } K_t &= P_{t|t-1} \times X_t' \times (X_t \times P_{t|t-1} \times X_t' + R)^{-1} \\ \theta_{t|t} &= \theta_{t|t-1} + K_t \times (Y_t - X_t \times \theta_{t|t-1}) \\ P_{t|t} &= (I - K_t \times X_t) \times P_{t|t-1}\end{aligned}$$

**Plain English:**  $K_t$  is the Kalman Gain — how much to trust new data vs prior belief. If  $R$  is small (data is precise),  $K$  is large — update substantially. If  $Q$  is small (parameters drift slowly), trust the prior more. The result: parameters update smoothly every period rather than jumping abruptly.

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## CHAPTER 19 — ADVANCED TOOLS

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[SLIDE 118] [NEW] *Visual: FAVAR equation system — two layers: factor model + VAR.*

**FAVAR: Factor-Augmented VAR — for when you have too many variables.**

Standard VAR struggles beyond 5-6 variables. FAVAR handles hundreds.

**Layer 1 — The Factor Model:**

$$X_t = \Lambda \cdot F_t + e_t$$

$X_t$  =  $N \times 1$  vector of many observables (N could be 100+)

$F_t$  =  $k \times 1$  vector of latent factors (k is small, say 3-5)

$\Lambda$  =  $N \times k$  factor loading matrix

$e_t$  =  $N \times 1$  idiosyncratic errors

Extract  $F_t$  via PCA on  $X_t$ . The first k principal components become your factors.

**Layer 2 — The VAR:**

$$\begin{bmatrix} F_t \\ y_t \end{bmatrix} = B(L) \times \begin{bmatrix} F_{t-1} \\ y_{t-1} \end{bmatrix} + \varepsilon_t$$

VAR runs on [Factors + Policy Variable]. Small system despite capturing 100+ variables.

**Estimation:** Two-step. PCA to extract factors, then VAR on  $[F_t, y_t]$ . Bernanke, Boivin, and Elias (2005) used this to study monetary policy transmission with hundreds of economic indicators.

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[SLIDE 119] [NEW] *Visual: Bayesian VAR — Minnesota Prior illustrated. Graph showing prior tightness by lag and variable type.*

**Bayesian VAR — Minnesota Prior — for large systems with limited data.**

For coefficient on lag j of variable k in equation for variable i:

**Own lags ( $k = i$ ):**

Prior mean: 1 if  $j=1$ , 0 if  $j>1$  (prior belief: AR(1) with coefficient 1)

Prior std:  $\lambda_1 \times \sigma_i / j$  (tightens for longer lags)

**Cross lags ( $k \neq i$ ):**

Prior mean: 0 (cross-variable effects shrink to zero)  
 Prior std:  $\lambda_1 \times \lambda_2 \times (\sigma_i / \sigma_k) / j$

$\lambda_1$  = overall tightness.  $\lambda_1 \rightarrow 0$ : hard-shrink to prior.  $\lambda_1 \rightarrow \infty$ : reduces to OLS.

### Posterior:

$$A_{\text{posterior}} = (X'X + V_{\text{prior}}^{-1})^{-1} \times (X'Y + V_{\text{prior}}^{-1} \times A_{\text{prior}})$$

Weighted average of OLS estimate and prior. More data  $\rightarrow$  posterior moves toward OLS.  
 Less data  $\rightarrow$  posterior stays near prior. This is how BVAR handles large systems gracefully.  
 Central banks worldwide use variants of this for macroeconomic forecasting.

[SLIDE 120] [NEW] *Visual: Historical Decomposition — actual series vs baseline forecast, decomposed into structural shock contributions.*

### Historical Decomposition — which shock caused what, when.

FEVD tells you average importance across time. Historical Decomposition tells you: "For August 2020 specifically — how much of Gold's observed move was caused by which structural shock?"

The formula:

$$Y_t - E_{\{t_0\}}[Y_t] = \sum_{s=t_0+1}^t \Psi_{\{t-s\}} \times P \times u_s$$

$Y_t$  = what actually happened  
 $E_{\{t_0\}}[Y_t]$  = baseline forecast made at time  $t_0$   
 $\Psi_{\{t-s\}}$  = VMA coefficient at horizon  $(t-s)$   
 $P$  = structural identification matrix  
 $u_s$  = estimated structural shock at time  $s$

Contribution of structural shock  $j$  from period  $s$  to variable  $i$  at time  $t$ :

$$HD(i, j, t, s) = (\Psi_{\{t-s\}} \times P)[i, j] \times u_{\{j, s\}}$$

In practice: baseline Gold forecast from January 2020. By March 2020, Gold had moved significantly. Historical decomposition tells you:

- 45% of that deviation: Global risk appetite shock (safe-haven flight)
- 35% of that deviation: Dollar policy shock (Fed rate cuts  $\rightarrow$  weaker dollar)
- 20% of that deviation: Gold-specific demand shock

Not averages. Exact attributions for those exact weeks. This is what gets published in central bank research and hedge fund attribution reports.

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## CLOSING — THE FULL MAP

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[SLIDE 121] *Visual: The complete VAR/VECM family detective board — all tools connected by strings and color-coded.*

Alright detectives. Let us step back and look at everything we built.

**The Core System:** VAR — the multi-variable engine. Every other model builds on it.

**Dynamics:** Granger Causality — who leads. IRF — how the system responds. OIRF — with structural ordering. GIRF — order-free. Lead-Lag — practical timing. FEVD — attribution of uncertainty.

**Long-Run Structure:** Cointegration — the long-run bond. VECM — respecting both short-run dynamics and long-run equilibrium.

**Causal Structure:** SVAR — recovering the true economic causal chain. Short-run restrictions. Long-run restrictions. Sign restrictions. Structural FEVD.

**Advanced Refinement:** Time-Varying VAR — drifting gradually. Markov-Switching VAR — discrete regime changes. And when  $K$  equals 1 — you are back to the standard VAR.

**Extended Toolkit:** FAVAR — for many variables. Bayesian VAR — for large, data-limited systems. Historical Decomposition — for specific event attribution.

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[SLIDE 122] *Visual: A scale — simplicity on left, realism on right. Each model placed on the scale.*

The final lesson.

Every step up this family adds one layer of realism and one layer of complexity. The question is always: does added complexity improve your answers enough to justify the cost in data, computation, and interpretability?

Standard VAR: simple, transparent, fast, great starting point. Markov-Switching VAR: realistic, captures crises, harder to estimate and explain.

The right model is not the most complex one. The right model is the one that answers your specific question with minimum necessary complexity.

Detectives do not bring in the forensic laboratory for a parking ticket.

But they also do not use only a flashlight when investigating a murder.

Know your question. Match your tool. Trust the data.

[SLIDE 123] *Visual: AlgoStalk outro card. Detective board aesthetic.*

That is the VAR/VECM family. Fully mapped. Every tool explained. Every calculation shown.

If this video helped you, hit the like button. Subscribe if you are new — we cover the entire algorithm universe one family at a time.

Comment below: which tool from today's video was the most surprising?

For me? The first time I saw GIRF, I thought: "Why did we ever bother with ordering if this existed?" The answer: because SVAR and structural identification REQUIRE ordering. GIRF is for exploration. OIRF is for structural claims. Both have their place.

Until the next case, detectives.

Stay sharp. Stay curious.

AlgoStalk. Out.

COMPLETE SLIDE SUMMARY

| Slide | Content                              | Status   |
|-------|--------------------------------------|----------|
| 1-5   | Hook — Option A                      | [HOOK]   |
| 6-9   | Introduction & Limitation            | Original |
| 10-17 | What is VAR?                         | Original |
| 18-20 | Correlated Errors — Full Calculation | [NEW]    |
| 21-27 | Stationarity & Lag Selection         | Original |
| 28-34 | Granger Causality                    | Original |
| 35-38 | IRF Concept & VMA Base               | Original |
| 39-44 | IRF Fully Calculated from Scratch    | [NEW]    |
| 45-51 | Cholesky — Why the Zero?             | [NEW]    |
| 52-56 | Orthogonalized IRF — Fully Worked    | [NEW]    |
| 57-59 | GIRF                                 | Original |
| 60-63 | Lead-Lag                             | Original |

| Slide   | Content                                              | Status           |
|---------|------------------------------------------------------|------------------|
| 64-68   | FEVD — Concept + Full Calculation                    | [NEW]            |
| 69-72   | FEVD Trading Application                             | [NEW]            |
| 73-76   | Covariance vs Correlation                            | Original         |
| 77-82   | Cointegration Concept                                | Original         |
| 79-80   | Gold-Silver Spread & Beta Calculation                | [NEW]            |
| 83-86   | Johansen Test — Full Procedure                       | [NEW]            |
| 87-90   | VECM — What is $\Delta Y$ ? Full Equation            | [NEW]            |
| 91      | Cointegrating Vector Calculation                     | [NEW]            |
| 92-94   | Alpha, Adjustment Speed, Decision Tree               | Original + [NEW] |
| 95-103  | SVAR & Identification                                | Original         |
| 98      | 9 Unknowns vs 6 Equations — Explicit                 | [NEW]            |
| 101     | Long-Run Restrictions (Blanchard-Quah)               | [NEW]            |
| 104-105 | Structural FEVD & Checkpoint                         | Original         |
| 106-109 | Time-Varying VAR                                     | Original         |
| 110-113 | Markov-Switching VAR                                 | Original         |
| 114-117 | Hamilton Filter & Kalman Filter — Step by Step       | [NEW]            |
| 118-120 | FAVAR, BVAR, Historical Decomposition with Equations | [NEW]            |
| 121-123 | Closing                                              | Original         |

## CHAPTER TIMESTAMPS

| Time  | Chapter                         |
|-------|---------------------------------|
| 0:00  | Hook                            |
| 2:30  | What is VAR?                    |
| 14:00 | Correlated Errors — Calculation |

| Time   | Chapter                                                      |
|--------|--------------------------------------------------------------|
| 18:30  | Stationarity & Lag Selection                                 |
| 24:00  | Granger Causality                                            |
| 33:00  | IRF — VMA Base + Full Calculation                            |
| 50:00  | Orthogonalized IRF — Cholesky + Zeros + Worked Example       |
| 68:00  | GIRF                                                         |
| 73:00  | Lead-Lag                                                     |
| 78:00  | FEVD — Full Calculation + Trading Application                |
| 98:00  | Covariance vs Correlation                                    |
| 102:00 | Cointegration — Spread, Beta Calculation, Johansen Procedure |
| 122:00 | VECM — $\Delta Y$ Explained + Beta Calculation               |
| 136:00 | SVAR — 9 Unknowns, Identification Methods incl. Long-Run     |
| 152:00 | Structural FEVD                                              |
| 157:00 | Time-Varying VAR                                             |
| 164:00 | Markov-Switching VAR + Hamilton & Kalman Filter              |
| 183:00 | FAVAR, BVAR, Historical Decomposition                        |
| 196:00 | Closing                                                      |